GENERAL PHARMACOLOGY

MBS230-P&T

UNIT 1.1.4

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LS 1.1.4- BIOAVAILABILITY & FIRST-PASS ELIMINATION

BIOAVAILABILITY AND FIRST PASS ELIMINATION

LS -1.1.4: LEARNING OBJECTIVES:

- Define bioavailability
- Explain the factors that affect bioavailability
- Describe first-pass elimination and explain its therapeutic implications

BIOAVAILABILITY AND FIRST PASS ELIMINATION

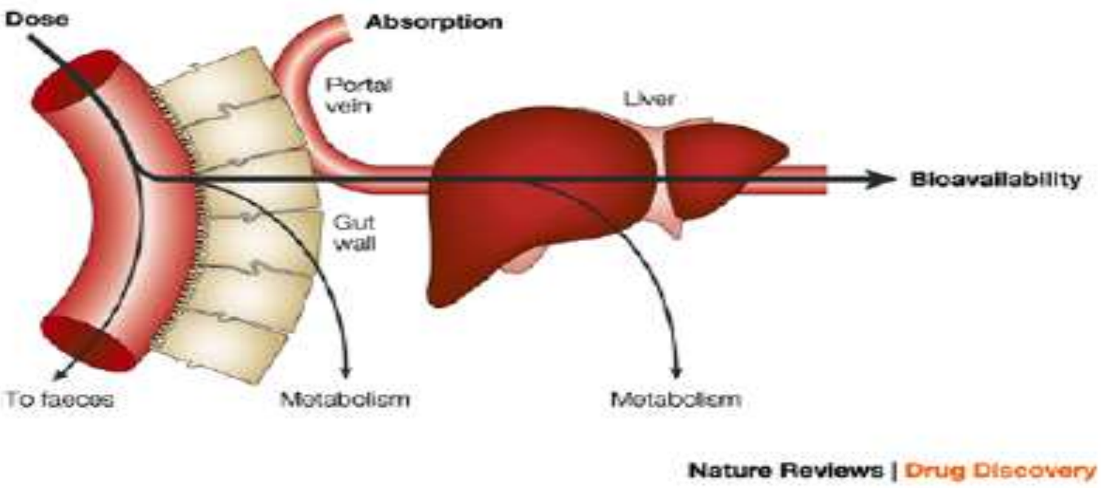
INTRODUCTION:

Bioavailability is the proportion of an administered dose of drug that reaches the systemic circulation in unchanged form

First pass elimination is the elimination of a drug during its passage from the site of absorption to the systemic circulation (also called pre-systemic elimination or first pass effect) and is an important feature of the oral route of administration

Bioavailability (F)

- The fraction of a dose that reaches the systemic circulation in a chemically unaltered form
- Fractional availability = F
- Quote as percentage or decimal e.g. 25% or 0.25
- Has no units



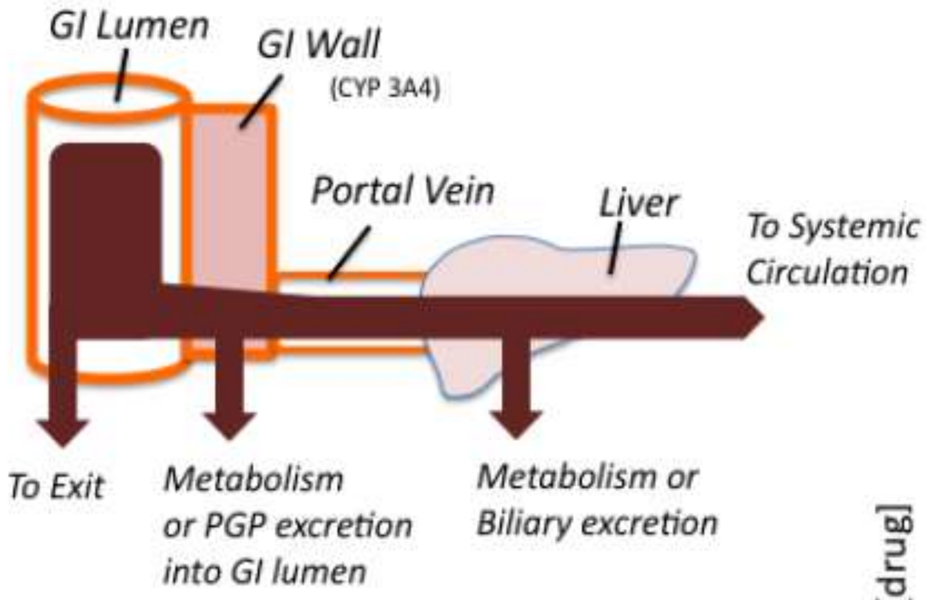
$$F = \frac{[AUC]_{po} * dose_{IV}}{[AUC]_{IV} * dose_{po}}$$

$$relative\ bioavailability = \frac{[AUC]_A * dose_B}{[AUC]_B * dose_A}$$

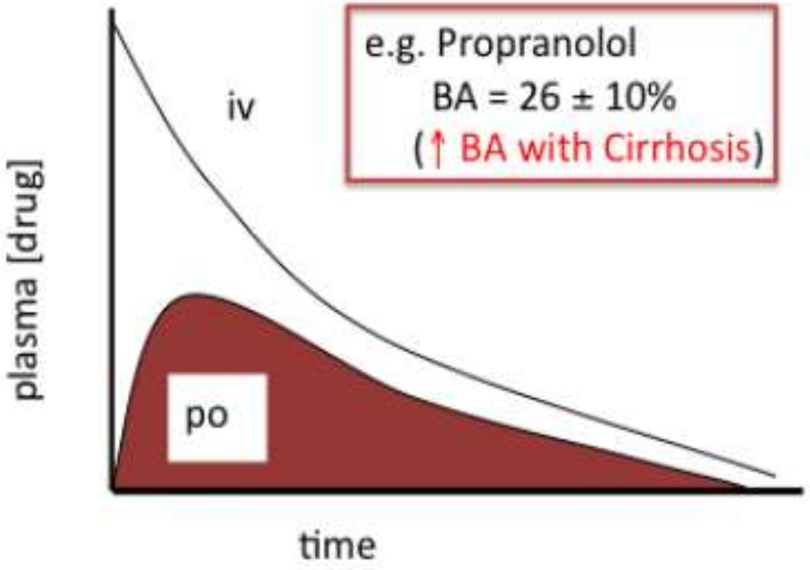
BIOAVAILABILITY (F)

- Bioavailability = amount of unchanged drug in the systemic circulation/total dose given
- Intravenous drugs have a bioavailability of 100%
- Drugs given by other routes vary in bioavailability
- Bioavailability affects: (1) The time of onset of drug effect (2) The time and magnitude of peak effect (3) The duration of effect i.e. how long drug response will last

Bioavailability (F)



$$BA = \frac{AUC_{po}}{AUC_{iv}} \times 100$$



BIOEQUIVALENCE

- Comparison of bioavailability of different formulations of the same drug is the study of bioequivalence
- Bioequivalence indicates that the drug products, when given to the same patient in the same dosage, regimen result in equivalent concentrations of drug in plasma and tissues

BIOEQUIVALENCE …. CONT'D

- Often oral formulations containing the same amount of drug from different manufacturers may result in different plasma concentrations, i.e. there is no bioequivalence among them
- Variations in bioavailability (non-equivalence) can result in toxicity or therapeutic failure for drugs with low therapeutic index. For such drugs, in a given patient, preparations from a single manufacturer should be used.

FACTORS THAT AFFECT BIOAVAILABILITY

- Route of drug administration
- Physicochemical properties of the drug
- Formulation
- First-pass elimination
- Concurrent drug therapy
- Food intake (for drugs administered orally)
- Disease states: bioavailability is reduced in diseases that cause malabsorption at the sites of drug absorption in the GIT, and when there is vomiting

FIRST-PASS ELIMINATION (PRE-SYSTEMIC ELIMINATION)

- First-pass elimination occurs when the drug is rapidly eliminated before it reaches the systemic circulation
- Following absorption across the gut wall, the portal blood delivers the drug to the liver prior to entry into the systemic circulation
- Metabolism may occur in the gut wall, portal venous blood or liver before the drug reaches the systemic circulation
- Most commonly, it is the liver that is responsible for metabolism before the drug reaches the systemic circulation. In addition the liver can excrete the drug into the bile.

FIRST-PASS ELIMINATION: THERAPEUTIC IMPLICATIONS

- A much larger dose of the drug is needed when it is given orally than when it is given by other routes due to reduced oral bioavailability
- Marked individual variations occur in the extent of first-pass elimination of a given drug resulting in unpredictability when such drugs are given orally
- First pass elimination is reduced in impaired liver function and in conditions in which there is shunting of blood into the systemic circulation without passing through the liver. Bioavailability in these cases is increased.

LEARNING OUTCOMES

Students must be able to comprehend:

- Definition of bioavailability and the factors that affect bioavailability
- First-pass elimination and its therapeutic implications

QUIZ

1. What does the term “bioavailability” mean?

a) Plasma protein binding degree of substance b) Permeability through the brain-blood barrier c) Fraction of an uncharged drug reaching the systemic circulation following any route administration d) Amount of a substance in urine relative to the initial doze

2. The reasons determing bioavailability are:

a) Rheological parameters of blood b) Amount of a substance obtained orally and quantity of intakes c) Extent of absorption and hepatic first-pass effect d) Glomerular filtration rate

3. Pick out the appropriate alimentary route of administration when passage of drugs through liver is minimized:

a) Oral b) Transdermal c) Rectal d) Intraduodenal

4. Which route of drug administration is most likely to lead to the first-pass effect?

a) Sublingual b) Oral c) Intravenous d) Intramuscular

END

Thanks for listening